

Runs of Ones in Binary Strings

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Update (March 2026): many more results about runs in binary strings can be found in our [latest paper](#). See also [“In for the long run”](#).

1 Introduction

A run of ones in a binary string is an uninterrupted sequence of ones flanked on each side either by a zero or by the start/end of the string (Mood’s criterion [1]). In the following, a run of ones will be simply referred to as a “run”.

What is the total number of runs of length i over all binary n -strings, or equivalently, if we draw binary n -strings uniformly at random, what is the expected number of runs of length i in a binary n -string?

This problem was previously solved by Sinha and Sinha [2] —in fact, these authors also solved a harder problem in [3] from which the solution to the problem addressed here can be produced.

Here we wish to show that the solution can be found in shorter and simpler ways than in [3] or [2] (see also [4]). We give three different solutions: two of them use elementary counting arguments (recursive and combinatorial solutions), while the third one is probabilistic.

2 Counting Runs Recursively

Let $r_n(i)$ be the total number of runs of length i over all binary n -strings.

First of all, let us get some visual intuition. In the diagrams below, for $n = 2, 3$ and 4 we list all binary n -strings and, right underneath, their runs “spectra” (i.e. the number of runs of lengths 1 to n found in each particular n -string). On the right we show $r_n(i)$.

- $n = 2$:

	0	0	1	1	
	0	1	0	1	
i					$r_2(i)$
1	0	1	1	0	2
2	0	0	0	1	1

- $n = 3$:

	0	0	0	0	1	1	1	1	
	0	0	1	1	0	0	1	1	
	0	1	0	1	0	1	0	1	
i									$r_3(i)$
1	0	1	1	0	1	2	0	0	5
2	0	0	0	1	0	0	1	0	2
3	0	0	0	0	0	0	0	1	1

- $n = 4$:

	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	
	0	0	0	0	1	1	1	1	0	0	0	0	1	1	1	1	
	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1	
	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	
i																	$r_4(i)$
1	0	1	1	0	1	2	0	0	1	2	2	1	0	1	0	0	12
2	0	0	0	1	0	0	1	0	0	0	0	1	1	1	0	0	5
3	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	2
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1

Assuming $n > 1$, it is trivial to see that

$$\begin{aligned} r_n(n) &= 1, \\ r_n(n-1) &= 2. \end{aligned} \tag{1}$$

This solves the problem for $n = 2$. Assuming $n > 2$, let us next see how, for $1 \leq i < n-1$, $r_n(i)$ can be recursively obtained from $r_{n-1}(i)$:

- On the one hand, consider the n -strings that start with a zero: these trivially contribute $r_{n-1}(i)$ runs to $r_n(i)$.
- On the other hand, consider the n -strings that start with a one. For $1 \leq i \leq n-1$, the 2^{n-i-1} n -strings that start with i ones followed by at least one zero add 2^{n-i-1} runs to $r_{n-1}(i)$ if $i < n-1$, and subtract 2^{n-i-1} runs from $r_{n-1}(i-1)$ if $i > 1$.

Thus, the n -strings that start with a one contribute $r_{n-1}(i) + 2^{n-i-1} - 2^{n-i-2} = r_{n-1}(i) + 2^{n-i-2}$ runs to $r_n(i)$.

So, collecting these two contributions we have that

$$r_n(i) = 2r_{n-1}(i) + 2^{n-i-2}. \quad (2)$$

Now, by using the expression above recursively k times, with $k < n - i$, we get

$$r_n(i) = 2^k r_{n-k}(i) + k 2^{n-i-2}. \quad (3)$$

When $k = n - i - 1$, we have that $r_{n-k}(i) = r_{i+1}(i) = 2$ because of (1). Thus, inputting this value of k in (3) we get the following explicit expression for the number of runs of ones of length i over all binary n -strings:

$$r_n(i) = (n - i + 3) 2^{n-i-2}, \quad (4)$$

where $1 \leq i < n - 1$. Incidentally, (4) is also valid when $i = n - 1$, i.e. it includes (1). Furthermore, we can see from (4) that $r_{n-1}(i - 1) = r_n(i)$ for $n \geq 2$, and so recurrence (2) can alternatively be expressed as a recurrence on i for a given n (rather than as a recurrence on n for a given i) as

$$r_n(i) = 2r_n(i + 1) + 2^{n-i-2}. \quad (5)$$

Also, the total number of runs over all n -strings is

$$t(n) = \sum_{i=1}^n r_n(i) = (n + 1) 2^{n-2}. \quad (6)$$

Recursion (2) was previously given in [2] and [4]. Also (4) and (6) were given in [2].

3 Counting Runs Using Combinatorics

Let us now give a way to obtain (6) directly by using combinatorics, i.e. without relying on (4). Consider the partitioning of an arbitrary n -string into p nonempty substrings, where $1 \leq p \leq n$. As shown in the example below, in which the n -string is represented by n asterisks and the p partitions by $p + 1$ vertical bars, we can put that partition into a bijection with p runs of ones from binary n -strings:

$$\begin{array}{c} |***|****|**|*****|\dots|*|**| \\ 111\ 0000\ 11\ 00000\ \dots\ 0\ 11 \\ 000\ 1111\ 00\ 11111\ \dots\ 1\ 00 \end{array}$$

The number of ways in which we can partition an n -string into p nonempty substrings is $\binom{n-1}{p-1}$. As we have p runs of ones associated to each of the $\binom{n-1}{p-1}$ possibilities, the total number of runs over all binary n -strings is

$$t(n) = \sum_{p=1}^n p \binom{n-1}{p-1} = (n + 1) 2^{n-2}. \quad (7)$$

Next, let us exploit the parallelism between expressions (4) and (6) to write $r_n(i)$ in a way that echoes (7). In order to do so, let us put (4) as the addition of the following two terms:

$$r_n(i) = (n - i + 1) 2^{n-i-2} + 2^{n-i-1}. \quad (8)$$

We can now see that the first term can be expanded using the same summation as in (7). Interestingly, by using $2^n = \sum_{k=0}^n \binom{n}{k}$, the second term can also be expanded into a summation with the same limits and over the same binomial coefficients as the first one, but without the p multiplication factors. Combining these two observations we can write (8) as follows:

$$\begin{aligned} r_n(i) &= \sum_{p=1}^{n-i} p \binom{n-i-1}{p-1} + \sum_{p=1}^{n-i} \binom{n-i-1}{p-1} \\ &= \sum_{p=1}^{n-i} (p+1) \binom{n-i-1}{p-1}. \end{aligned} \quad (9)$$

Expression (9) literally tells us that there is a one-to-one correspondence between a partitioning of an arbitrary $(n-i)$ -string into p nonempty substrings and $p+1$ runs of length i from binary n -strings. This provides a simple combinatorial approach to obtaining (4).

In order to see how the bijection implied by (9) works we just need to place a run of ones of length i at each of the $p+1$ possible positions in the partitioning (i.e. the $p-1$ divisions plus the start and the end of the string) and then complete a binary n -string with alternating runs of ones and zeros having lengths determined by the partitioning.

For example, let us take $n = 4$ and $i = 1$ and explicitly illustrate the correspondence between each of the possible partitionings of an arbitrary 3-string “* * *” into p nonempty substrings and $p+1$ runs of length 1 in binary 4-strings:

- $p = 1 \rightarrow \binom{2}{0} = 1$ partitioning associated with 2 runs of length 1

***	1000
***	0001

- $p = 2 \rightarrow \binom{2}{1} = 2$ partitionings, each associated with 3 runs of length 1:

* **	
* **	1011
* **	0100
* **	1001

** *	
** *	1001
** *	0010
** *	1101

- $p = 3 \rightarrow \binom{2}{2} = 1$ partitioning, associated with 4 runs of length 1:

* * *	
* * *	1010
* * *	0101
* * *	1010
* * *	0101

If $n = 4$ and $i = 2$ we look at the partitionings of “**” into p nonempty sub-strings:

- $p = 1 \rightarrow \binom{1}{0} = 1$ partitioning associated with 2 runs of length 2

**	
**	1100
**	0011

- $p = 2 \rightarrow \binom{1}{1} = 1$ partitioning associated with 3 runs of length 2

* *	
* *	1101
* *	0110
* *	1011

Finally, if we let the sum in (9) start at $p = 0$ rather than $p = 1$, then (9) also includes the case¹ $i = n$, whereas (4) is only valid for $i \leq n - 1$.

4 Counting Runs Probabilistically

Assume that the binary n -strings are generated uniformly at random, and define n indicator random variables $S_1, \dots, S_n$ such that $S_k = 1$ if a run of ones starts at the k -th position of the string and 0 otherwise. There are two cases for the probability of S_k :

$$\Pr(S_k = 1) = \begin{cases} \frac{1}{2} & \text{if } k = 1 \\ \frac{1}{4} & \text{if } 2 \leq k \leq n \end{cases}, \quad (10)$$

¹Using $\binom{-1}{-1} = 1$ [5] and $\binom{t}{-1} = 0$ for all nonnegative integers t .

as if a run starts at position $k > 1$ then the previous bit must be zero. The total number of runs is thus given by r.v. $T = \sum_{k=1}^n S_k$, whose expectation $E(T) = \sum_{k=1}^n E(S_k) = \sum_{k=1}^n \Pr(S_k = 1)$ is

$$E(T) = (n+1)2^{-2}. \quad (11)$$

The rationale above was given by Alex Proscurin in [Expected value of a run of a random bitstring](#). As we can see, (11) is the normalisation of (6), i.e. $E(T) = t(n)/2^n$.

Let us extend the procedure above and define indicator r.v.'s $R_k^{(i)}$ such that $\Pr(R_k^{(i)} = 1)$ is the probability that a run of length i starts at position k . We have that

$$\Pr(R_k^{(i)} = 1) = \Pr(R_k^{(i)} = 1 | S_k = 1) \Pr(S_k = 1), \quad (12)$$

because $\Pr(R_k^{(i)} = 1 | S_k = 0) = 0$. As for the conditional probabilities in (12), there are three cases:

$$\Pr(R_k^{(i)} = 1 | S_k = 1) = \begin{cases} 2^{-i} & \text{if } 1 \leq k \leq n-i \\ 2^{-(i-1)} & \text{if } k = n-i+1 \\ 0 & \text{if } n-i+1 < k \leq n \end{cases}. \quad (13)$$

The first two cases are just the application of the geometric distribution with parameter $1/2$, noting that in the second case the run does not need to be finished by a zero. Now, the number of runs of length i is given by r.v. $T^{(i)} = \sum_{k=1}^n R_k^{(i)}$, whose expectation $E(T^{(i)}) = \sum_{k=1}^n E(R_k^{(i)}) = \sum_{k=1}^n \Pr(R_k^{(i)} = 1)$ is

$$E(T^{(i)}) = (n-i+3)2^{-i-2}. \quad (14)$$

Again, we have that (14) is the normalisation of (4), i.e. $E(T^{(i)}) = r_n(i)/2^n$.

The total fraction of runs of length i is

$$f_n(i) = \frac{r_n(i)}{t(n)} = \frac{n-i+3}{n+1} 2^{-i} \quad (15)$$

for $1 \leq i \leq n-1$, whereas $f_n(n) = 1/t(n)$. Interestingly, $f_n(2) = 1/4$ for any $n > 2$. For large n , (15) should approximate the fraction of runs of length i in one single binary n -string drawn uniformly at random, in which case $f_n(i) \approx 2^{-i}$.

5 Connections with Compositions, and Relationship to OEIS Sequences

Sequence [A045623](#) from OEIS [6] (number of 1's in all compositions of $j+1$) is defined by $a(j) = (j+3)2^{j-2}$ for $j \geq 1$ and $a(0) = 1$. Observing (4), $r_n(i) = a(n-i)$.

Also, sequence [A001792](#) (number of parts in all compositions of $j + 1$) is defined by $b(j) = (j + 2) 2^{j-1}$. From (6), $t(n) = b(n - 1)$.

Similarly, the equivalent of expression (7) and expression (9) are given in [A045623](#) and [A001792](#), respectively.

These connections are due to the bijection between runs in binary n -strings and parts of the compositions of n , which is easy to observe when the compositions are put as the different ways in which the sum $1 + 1 + \dots + 1 = n$ can be partitioned into nonzero parts (see diagram at start of Section 3). Therefore, the runs in all binary n -strings are in a one-to-one correspondence with the parts in all compositions of n , which is $t(n)$ in both cases. Similarly, the runs of length 1 in all binary n -strings are in a one-to-one correspondence with the 1's in all compositions of n , which is $r_n(1)$ in both cases.

After noticing the connection between runs in binary strings and compositions, we found that the gist of the counting argument that we used in Section 2 was previously given by Mike Earnest in [Number of parts equal to 1 in all compositions of \$n\$](#) . Also, the argument to get (7) must be well known in the compositions problem.

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